RQ Answers

omitted

2024-04-05

Data overview

```
glimpse(all_data)
```

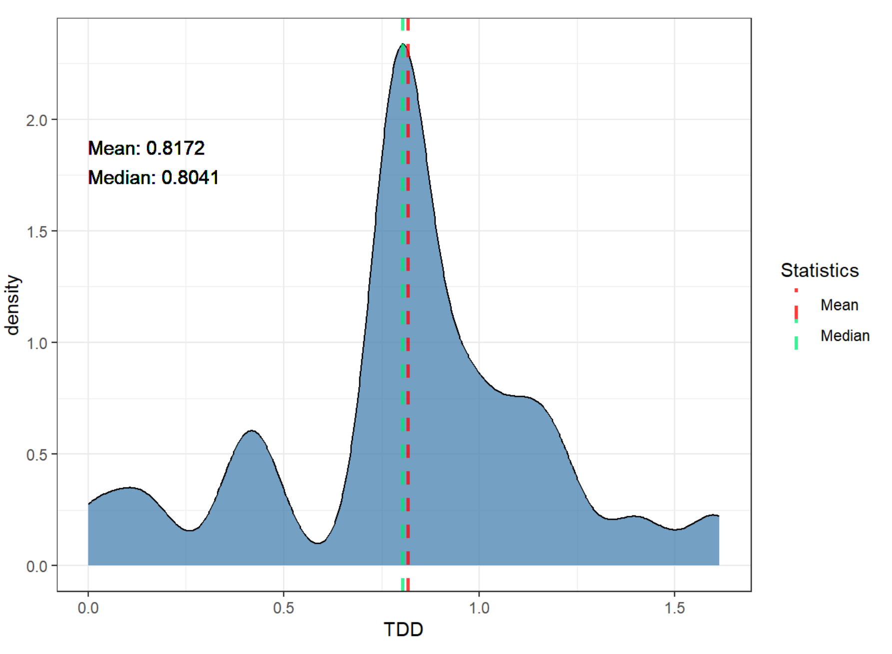
```
## Rows: 7,328
## Columns: 11
## $ repo              <chr> "commons-io", "commons-io", "commons-io", "commons…
## $ pr_number         <int> 177, 177, 177, 177, 177, 177, 177, 177, 177, 177, …
## $ rule              <chr> "java:S1192", "java:S4719", "java:S4719", "java:S4…
## $ severity          <chr> "CRITICAL", "MINOR", "MINOR", "MINOR", "MINOR", "M…
## $ file              <chr> "src/test/java/org/apache/commons/io/IOUtilsCopyTe…
## $ type              <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status            <chr> "CLOSED", "CLOSED", "CLOSED", "CLOSED", "CLOSED", …
## $ debt              <int> 42, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1…
## $ ncloc_affected_file <int> 306, 306, 306, 306, 306, 306, 306, 306, 306, 306, …
## $ complexity        <int> 41, 41, 41, 41, 41, 41, 41, 41, 41, 41, 41, 41, 41…
## $ origin            <chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P…
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 52 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 52 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

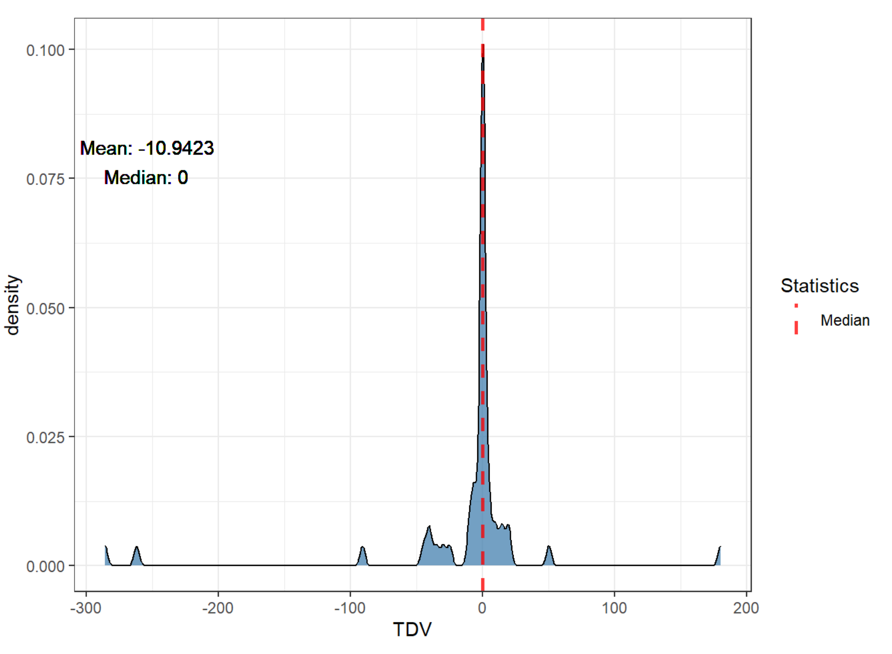


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 52 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 52 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 × 4
##   repo       tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>                    <dbl>             <dbl>                 <dbl>
## 1 commons-io                28.8                50                  21.2
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                      <dbl>
## 1                     28.8                     50                       21.2
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 × 3
##   repo       preexisting_tdv_equal_to_zero preexisting_tdv_greater_than_zero
##   <chr>                              <dbl>                             <dbl>
## 1 commons-io                          66.7                              33.3
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                          66.7                                   33.3
```

Percentages for new TDV by repo

```
## # A tibble: 1 × 3
## # Groups:   repo [1]
##   repo       new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                      <dbl>                     <dbl>
## 1 commons-io                  88.9                      11.1
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                       88.9                           11.1
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule i, its $PositionScore_i$ will be the sum of its position across all k repositiries.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##     rank `commons-io`
##    <int> <chr>
##  1     1 java:S4719  -  98
##  2     2 java:S1192  -  24
##  3     3 java:S4042  -  11
##  4     4 java:S899  -  11
##  5     5 java:S2129  -  7
##  6     6 java:S2293  -  6
##  7     7 java:S1874  -  6
##  8     8 java:S1481  -  5
##  9     9 java:S125  -  4
## 10    10 java:S1854  -  3
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S4719            122
##  2 java:S1192            123
##  3 java:S4042            124
##  4 java:S899             125
##  5 java:S2129            126
##  6 java:S1874            127
##  7 java:S2293            128
##  8 java:S1481            129
##  9 java:S125             130
## 10 java:S106             131
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##     rank `commons-io`
##    <int> <chr>
##  1     1 java:S1192  -  2061
##  2     2 java:S100  -  1326
##  3     3 java:S899  -  701
##  4     4 java:S4042  -  544
##  5     5 java:S4719  -  398
##  6     6 java:S2184  -  327
##  7     7 java:S117  -  253
##  8     8 java:S108  -  243
##  9     9 java:S3878  -  182
## 10    10 java:S1874  -  153
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S1192            122
##  2 java:S100             123
##  3 java:S899             124
##  4 java:S4042            125
##  5 java:S4719            126
##  6 java:S2184            127
##  7 java:S117             128
##  8 java:S108             129
##  9 java:S3878            130
## 10 java:S1874            131
```

Outliers

Higher TDD

```
## # A tibble: 52 × 5
##    repo       pr_number total_debt total_ncloc   tdd
##    <chr>          <int>      <int>       <int> <dbl>
##  1 commons-io       375        922         571  1.61
##  2 commons-io       535        415         261  1.59
##  3 commons-io       580        140          98  1.43
##  4 commons-io       530        530         385  1.38
##  5 commons-io       201       2232        1807  1.24
##  6 commons-io       450        185         154  1.20
##  7 commons-io        79       1165         992  1.17
##  8 commons-io       179       1163        1000  1.16
##  9 commons-io       526        109          95  1.15
## 10 commons-io        19       1050         939  1.12
## # ℹ 42 more rows
```

Higher TDV

```
## # A tibble: 52 × 3
##    pr_number repo         tdv
##        <int> <chr>      <int>
##  1       450 commons-io   180
##  2       580 commons-io    50
##  3       525 commons-io    20
##  4       558 commons-io    19
##  5       452 commons-io    15
##  6       570 commons-io    14
##  7        44 commons-io    10
##  8       564 commons-io     9
##  9       194 commons-io     5
## 10       325 commons-io     5
## # ℹ 42 more rows
```

Lower TDV

```
## # A tibble: 52 × 3
##    pr_number repo         tdv
##        <int> <chr>      <int>
##  1       535 commons-io  -286
##  2       177 commons-io  -262
##  3       475 commons-io   -91
##  4       531 commons-io   -45
##  5       267 commons-io   -41
##  6       530 commons-io   -40
##  7       526 commons-io   -35
##  8       562 commons-io   -30
##  9       577 commons-io   -25
## 10       340 commons-io   -10
## # ℹ 42 more rows
```

Extra

Number of PRs with pre-existing TD

```
##    n
## 1 51
```

Number of java:S1192 issues that affect test code

```
##      n
## 1 2085
```

Number of java:S1192 issues that affect not test code

```
##   n
## 1 0
```

Summary NCLOC

```
##      ncloc
##  Min.   :  58.0
##  1st Qu.: 247.2
##  Median :1015.5
##  Mean   :1696.2
##  3rd Qu.:2547.5
##  Max.   :9408.0
```